

Longitudinal Gaps and Readout Connectivity in a Pilot Generative Model of Zero-Degree Calorimeter Showers

Julian Juan

juliansjuan08@gmail.com

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Abstract

We examine longitudinal gaps and graph fragmentation in a hierarchical single-neutron generator for a 6,790-channel simulation geometry of the ePIC Zero-Degree Calorimeter. The target is raw deposited energy, without digitization or an analysis threshold. One checkpoint trained on 26,624 showers is compared with Geant4 at 10,000 matched incident conditions spanning 50–250 GeV kinetic energy. Among nonempty showers, the generated sample has 0.25 more active layers on average, but its last active layer lies 4.42 layers farther downstream and its occupied span contains 3.95 more inactive layers. The mean graph-component count increases by 35.97. Since each occupied-layer run needs at least one component, we subtract that unavoidable contribution: the remaining component-count difference is at least 29.92, or 83.2% of the observed excess. Thus empty-layer separations alone do not account for the graph fragmentation. These descriptive results concern one training seed and a repeatedly inspected development bank. Variation across fits, detector thresholds, and adjacency definitions has not been tested. Conditional activity factorization and fixed-count channel sampling provide specific hypotheses for follow-up tests; the present comparison does not establish calibrated detector performance.

1 Introduction

Detailed particle transport provides the reference calorimeter showers but can be computationally costly. Before detector reconstruction is considered, a surrogate can be tested against the spatial organization of its stored simulation deposits. For sparse showers, the occupied layers and channels, their correlations, and their spatial organization are part of that target. Here, support means the set of channels with positive deposit; occupancy counts its members.

Generative calorimeter studies have used normalizing flows, diffusion, and conditional flow matching with regular-grid, point-cloud, or graph representations [1–5]. CaloChallenge assesses complementary energy, shape, classifier, and computational-cost observables [6, 7]. Work on the ALICE Zero-Degree Calorimeter (ZDC) has explored variational autoencoders, adversarial models, diffusion, flow matching, and mixtures of generative experts [8–12]. Recent ZDC flow studies address output variability [13] and neighboring-channel co-activation through a conditional Jaccard error [14]. Those models use different detector representations and targets, so their reported numerical fidelities are not direct baselines for the raw channel energies studied here.

We present a diagnostic study of a hierarchical ZDC shower generator that resolves the readout into total deposit, layer activity, and channel support. The central question is whether agreement in mean response and occupancy also captures the longitudinal organization of a shower. In the pilot comparison studied here, similar means coexist with later deposits and substantially more inactive layers inside the occupied span. We derive an aggregate bound showing that most of the component-count excess remains within occupied-layer runs. This graph diagnostic complements co-activation tests; the general need for correlation-sensitive validation is established in the cited literature.

The analysis uses one trained checkpoint and a 10,000-condition development bank drawn from validation. This is a diagnostic case study; no architecture-superiority comparison is performed.

2 Simulation target and study population

2.1 Readout target and conditioning

The event condition is the incident-neutron four-vector $p^\mu = (E, p_x, p_y, p_z)$, expressed in GeV. The five network inputs are

$$c = (\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E/\text{GeV})), \quad \mathbf{u} = \frac{\mathbf{p}}{\max(\|\mathbf{p}\|, 10^{-12} \text{ GeV})}, \quad (1)$$

where $K_{\text{inc}} = \max(E - m_n, 0)$ and $m_n = 0.93956542052 \text{ GeV}$. The direction features are zero at zero momentum. For $E \geq m_n$, the two energy features carry the same physical information because $E = K_{\text{inc}} + m_n$; the denominator clamp is numerical.

The target $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{6790}$ is the raw, nondigitized energy deposited in each valid readout channel, with repeated hits to one ID summed in preprocessing. This is our data convention; Geant4 supplies the transport [15]. Its event total is a sum of stored readout deposits, not the incident energy or the total energy lost in detector material. No analysis threshold or additional time selection is applied; the production integration window is unknown. We call an event with zero total readout deposit an empty shower.

2.2 Geometry and graph representation

This is an ePIC ZDC simulation [16], with an unrecorded production geometry tag. Its stored readout contains 400 electromagnetic-calorimeter (ECAL) channels in layer 0 and 6,390 hadronic-calorimeter (HCAL) channels in layers 1–64: 100 channels in each of layers 1–63 and 90 in layer 64. Of the HCAL IDs, 2,400 combine deposits from two to four stable physical positions into one readout channel (ganging). We represent a ganged channel by the unweighted centroid of those positions (Fig. 1), so the spatial resolution is set by the readout ID. The production material specification, Geant4 version, physics list, cuts, field, coordinate frame and four-vector recording surface are unverified. Layer indices are not depths in radiation or interaction lengths. The raw ECAL+HCAL sum is uncalibrated; equivalence to a current detector geometry is not established.

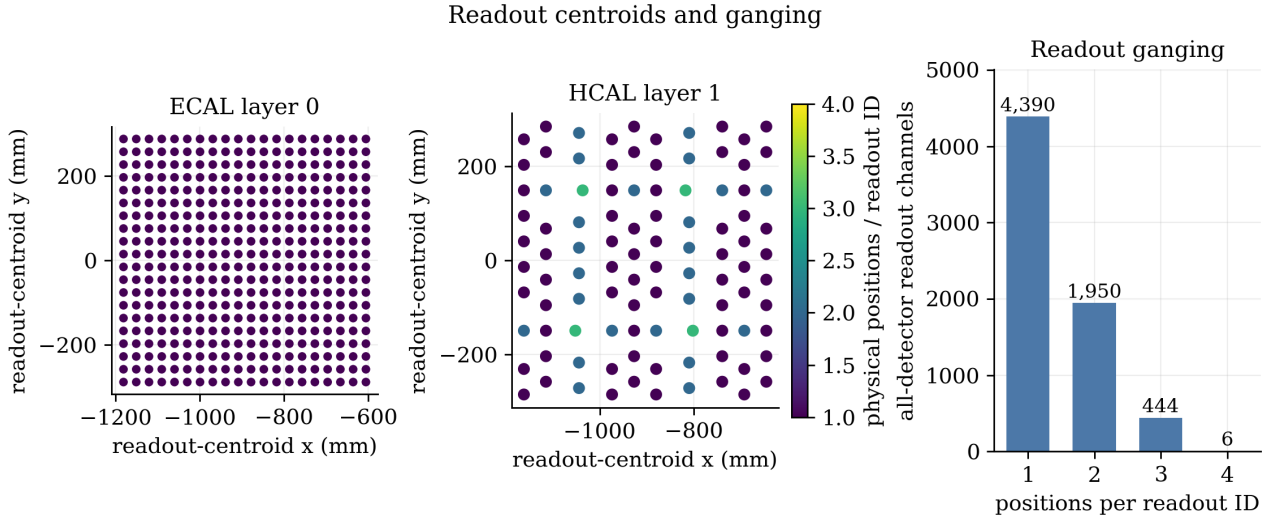


Figure 1: Frozen readout geometry. The left panels show ECAL layer 0 and HCAL layer 1. The ganging histogram covers all 6,790 readout channels; its first bar contains 400 ECAL and 3,990 HCAL channels. These centroids define the spatial representation for the model and connectivity diagnostic.

The fixed graph gives each channel eight nearest-neighbor links within its layer. For each channel in layers 0–63, it also selects four nearest centroids in the next layer, with each selected longitudinal pair stored in both directions. The resulting directed edge set contains $8 \times 6790 = 54,320$ lateral and $2 \times 4 \times (400 + 63 \times 100) = 53,600$ longitudinal edges, or 107,920 in total. The generator and connectivity diagnostic share this graph, which represents adjacency through readout-centroid distances.

2.3 Training and evaluation populations

The preparation record contains 764,940 showers over 0–300 GeV. A deterministic event-hash split assigns 612,482 to canonical training, 76,158 to validation, and 76,300 to the nominal test partition. The selected checkpoint instead used a pilot subset of 26,624 training and 6,656 validation events; the pilot training population is 4.35% of canonical training.

The primary shower comparison uses 10,000 incident conditions drawn from canonical validation with $50 \leq K_{\text{inc}} \leq 250$ GeV. We refer to this repeatedly inspected validation subset as the development bank D_{dev} . For each four-vector, we compare one stored Geant4 shower with one independently sampled model shower. The pair shares its incident condition; its stochastic shower histories are independent. The development bank’s overlap with the 6,656-event pilot-validation subset cannot be reconstructed from the released aggregates, although the recorded split cross-tabulation places pilot training and validation events within their corresponding canonical partitions. No nominal test event is used here. One draw per condition tests bank-level distributions, not the full shower distribution at a fixed condition.

3 Generator and training procedure

The generator builds a shower at three physical scales (Fig. 2). It first draws whether any readout deposit is visible and how much energy is deposited in total. It next determines which layers are active and allocates that total among them. Finally, within each active layer it draws a channel count, selects those channels, and divides the layer budget among them. A downstream stage uses the sampled outputs of upstream stages, so agreement in a final observable depends on the full cascade rather than on its last network alone.

The model has 2,082,507 trainable parameters. A two-block residual multilayer perceptron encodes the five inputs in Eq. (1) into a 128-component condition vector h_c . Discrete heads generate visibility, layer activity, channel counts, and support; two conditional flow-matching models generate the continuous layer and channel energy shares from independent Gaussian draws. There is no learned shower encoder or common random event latent. At inference, the incident four-vector is the only event input.

Below, $\ell = 0, \dots, 64$ indexes layers, i indexes valid channels in layer ℓ , and $Y_{\ell i}$ is a stored deposit. A channel is active if $Y_{\ell i} > 0$ and a layer if its budget $B_\ell = \sum_i Y_{\ell i} > 0$. The event total is $T = \sum_\ell B_\ell$.

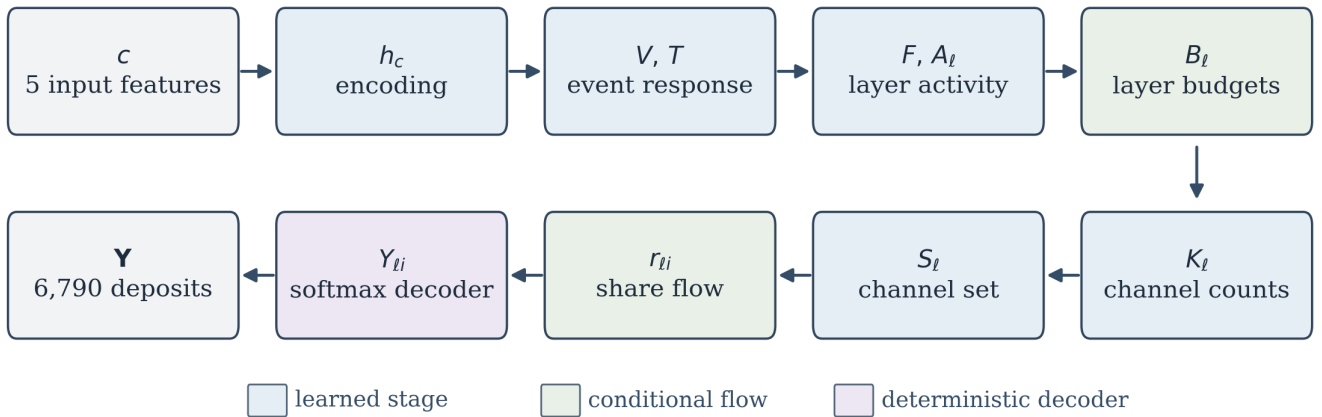


Figure 2: Generation proceeds left to right across the top row and back across the bottom. Visibility V , total T , first-layer F , activity A_ℓ , budgets B_ℓ , counts K_ℓ and support S_ℓ precede channel-energy logits $r_{\ell i}$ and deposits $Y_{\ell i}$. Green stages are conditional flows; purple denotes deterministic decoding. Arrows show generation order, not all conditioning connections or transport. Training supplies reference upstream states; sampling supplies generated states.

3.1 Total deposit and longitudinal allocation

The response stage separates empty readout from the size of a nonempty deposit. A Bernoulli head samples a provisional visibility flag V_0 , while a four-component Gaussian mixture describes the transformed total q_T . The Bernoulli logit a_θ , mixture means μ_m , positive widths σ_m , and softmax weights ω_m depend on h_c ; θ denotes the learned parameters and σ the logistic function:

$$\begin{aligned} \Pr(V_0 = 1 \mid h_c) &= \sigma(a_\theta(h_c)), & q_T &= \log(1 + T/10 \text{ GeV}), \\ p_\theta(q \mid h_c) &= \sum_{m=1}^4 \omega_m(h_c) \mathcal{N}(q; \mu_m(h_c), \sigma_m^2(h_c)). \end{aligned} \quad (2)$$

A draw q is mapped back to deposited-energy units, clipped at zero, and capped at the training-derived value $C(K_{\text{inc}})$. The final visibility flag also requires positive incident kinetic energy and a positive candidate deposit:

$$\begin{aligned} C(K_{\text{inc}}) &= \min\{0.630110 \max(K_{\text{inc}}, 1 \text{ GeV}), 61.2338 \text{ GeV}\}, \\ T_{\text{cand}} &= \min\{10 \text{ GeV} \max(e^q - 1, 0), C(K_{\text{inc}})\}, \\ V &= V_0 \mathbf{1}_{K_{\text{inc}} > 0} \mathbf{1}_{T_{\text{cand}} > 0}, & T &= VT_{\text{cand}}. \end{aligned} \quad (3)$$

The cap saturates at 61.2338 GeV above $K_{\text{inc}} \approx 97.2 \text{ GeV}$. Its training-derived tail bound is a numerical safeguard, not a sampling fraction or a mean-response prediction. Clamping a nonpositive inverse-transform value to zero can produce an empty shower even when $V_0 = 1$.

For a visible event, the model draws the first active layer $F \sim \text{Cat}[\text{softmax } f_\theta(h_c, \log(1 + T/\text{GeV}))]$ from 65 categorical probabilities. It forces layer F active and all earlier layers inactive. Separate Bernoulli draws set the later indicators A_ℓ ; conditional on (h_c, T, F) , their joint probability factorizes as

$$p(\mathbf{A} \mid h_c, T, F) = \prod_{\ell > F}^{64} p(A_\ell \mid h_c, T, F, \ell), \quad A_F = 1. \quad (4)$$

The activity mask may therefore contain empty layers between active ones. The budget flow subsequently distributes T only across active layers. A star denotes a reference-derived quantity. For a reference shower, let $B_\ell^* = \sum_i Y_{\ell i}^*$, $T^* = \sum_\ell B_\ell^*$, and $A_\ell^* = \mathbf{1}_{B_\ell^* > 0}$. The target for a nonempty event is a centered log fraction on those layers:

$$a_\ell^* = \log \max\left(\frac{B_\ell^*}{\max(T^*, 10^{-12} \text{ GeV})}, 10^{-8}\right), \quad (x_1)_\ell = A_\ell^* \left(a_\ell^* - \frac{\sum_m A_m^* a_m^*}{\max(1, \sum_m A_m^*)}\right). \quad (5)$$

For an empty reference shower, the target and mask are zero. The centering removes a common log offset, leaving the relative layer shares that enter the later softmax. Floors of 10^{-8} and 10^{-12} GeV make the logarithm and division finite; they are numerical conventions, not detector thresholds. Fractions below the floor are modified before centering, so these targets do not preserve arbitrarily small shares exactly.

Both energy-allocation flows learn a velocity field that transports Gaussian noise toward the distribution of reference log fractions, conditioned on the incident particle and upstream shower state. Flow time is an artificial interpolation coordinate, independent of detector time. Straight-path conditional flow matching trains this field by regressing the displacement along noise-to-target interpolations [17]. For each training example, x_0 is standard-normal noise on the active coordinates, x_1 the reference-derived target, and t is uniform on $[0, 1]$. A mask $M_{bj} \in \{0, 1\}$ identifies active coordinate j of minibatch example b :

$$x_t = (1 - t)x_0 + tx_1, \quad \mathcal{L}_{\text{FM}} = \frac{\sum_{b,j} M_{bj} [v_\theta(x_t, t; \text{ctx})_{bj} - (x_1 - x_0)_{bj}]^2}{\max(1, \sum_{b,j} M_{bj})}. \quad (6)$$

The velocity field v_θ predicts the straight-path displacement $x_1 - x_0$ from the interpolated state x_t , time, and context. Inactive coordinates do not contribute to the masked squared-error loss.

During generation, both flows start from fresh Gaussian noise and use eight masked velocity updates. The context contains outputs sampled by earlier stages, whereas training supplies their reference values. For an active-layer or selected-channel mask M , the implemented rule is

$$z_0 = M \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad z_{n+1} = M \odot \left[z_n + \frac{1}{8} v_\theta \left(z_n, \frac{n+1/2}{8}; \text{ctx} \right) \right], \quad n = 0, \dots, 7. \quad (7)$$

Here $\odot$ denotes elementwise multiplication. Each update evaluates the velocity at the current state z_n and the midpoint *time* $(n + 1/2)/8$, then resets inactive coordinates to zero. It is a first-order explicit update with midpoint-time evaluation, not the midpoint Runge–Kutta method, which also advances the state to the midpoint. For a visible event, the resulting active-layer logits are normalized to fractions π_ℓ , giving budgets

$$\pi_\ell = \frac{A_\ell e^{z_{8,\ell}}}{\sum_{m:A_m=1} e^{z_{8,m}}}, \quad B_\ell = T\pi_\ell, \quad \sum_{\ell=0}^{64} B_\ell = T. \quad (8)$$

Only active layers enter the softmax denominator; their budgets sum to the sampled total and inactive layers receive zero. The layer-flow network uses a 128-component representation and two bidirectional transformer layers with four attention heads. Its context includes h_c , the total T , layer identity, the activity mask, and a sinusoidal flow-time encoding. Bidirectional attention lets each layer use information from all other layers; it is a statistical dependency, not backward particle transport.

3.2 Channel count, support, and deposited energy

Given (h_c, B_ℓ, A_ℓ) and a learned layer identifier, a categorical head draws the number K_ℓ of active channels. A feasibility mask sets $K_\ell = 0$ for inactive layers and restricts active layers to $1 \leq K_\ell \leq N_\ell$, where N_ℓ is the number of valid channels. This fixes occupancy, not the positions of selected channels or a minimum deposit per channel.

The support network uses channel geometry and the sampled layer state to score candidate channels. Its inputs combine standardized (x, y, z) centroids, normalized layer index, ECAL/HCAL indicators, h_c , B_ℓ , and the requested fraction K_ℓ/N_ℓ . Three graph blocks of width 96 propagate information along the fixed directed edge set $\mathcal{E}$. Each block forms messages from neighboring readouts and uses their sum to update the receiving channel. If $h_i^{(r)}$ is the representation of channel i entering block r , and e_{ji} contains five edge inputs (three coordinate differences, distance, and edge type), one block computes

$$m_i^{(r)} = \sum_{j:(j,i) \in \mathcal{E}} \phi_r(h_j^{(r)}, h_i^{(r)}, e_{ji}), \quad h_i^{(r+1)} = h_i^{(r)} + \psi_r(h_i^{(r)}, m_i^{(r)}). \quad (9)$$

Here ϕ_r computes each message and ψ_r computes the residual update. Layerwise averages of the resulting representations pass through a two-layer, four-head bidirectional transformer; its outputs are broadcast back to channels before their final scores $s_{\ell i}$ are formed. A graph message can make a score sensitive to nearby readouts, but selection remains a separate stochastic step. For valid-channel set $\mathcal{N}_\ell$, the sampler chooses

$$S_\ell = \{i \in \mathcal{N}_\ell : \text{rank}_{\mathcal{N}_\ell}(s_{\ell i}/\tau + g_{\ell i}) \leq K_\ell\}, \quad (10)$$

Rank one denotes the largest noisy score. The sampler uses temperature $\tau = 1$ and independent Gumbel perturbations $g_{\ell i}$ obtained from uniform draws clipped to $[10^{-7}, 1 - 10^{-7}]$ [18]. Selecting the top K_ℓ enforces the requested count exactly, but imposes no adjacency or connectivity constraint.

A second flow divides each layer budget among its selected channels. In training, $S_\ell^\star$ denotes the positive-deposit reference channels and $K_\ell^\star = |S_\ell^\star|$. For $K_\ell^\star > 0$, its target is the centered log fraction of each channel’s layer budget:

$$a_{\ell i}^\star = \log \max \left(\frac{Y_{\ell i}^\star}{\max(B_\ell^\star, 10^{-12} \text{ GeV})}, 10^{-8} \right), \quad (x_1)_{\ell i} = \mathbf{1}_{i \in S_\ell^\star} \left(a_{\ell i}^\star - \frac{1}{K_\ell^\star} \sum_{j \in S_\ell^\star} a_{\ell j}^\star \right). \quad (11)$$

Here x_1 refers to the channel-flow target rather than the layer-flow target. The centering preserves the floored relative shares. An inactive layer has zero target coordinates. This flow uses Eqs. (6) and (7) with the reference channel set during training and the sampled set during generation.

The channel-flow velocity network follows the support network’s graph and layer-context design, with the flow state and a sinusoidal time encoding supplied at each channel. Unselected channels are masked before and after every graph block. Eight updates yield logits $r_{\ell i}$ for selected channels; a deterministic softmax then assigns the energy

$$Y_{\ell i} = \begin{cases} B_\ell \frac{\exp r_{\ell i}}{\sum_{j \in S_\ell} \exp r_{\ell j}}, & i \in S_\ell, \\ 0, & i \notin S_\ell, \end{cases} \quad (12)$$

Consequently, in exact arithmetic, $|S_\ell| = K_\ell$, $Y_{\ell i} \geq 0$, $\sum_i Y_{\ell i} = B_\ell$, and $\sum_{\ell i} Y_{\ell i} = T$. These are accounting identities for the model’s sampled readout budget.

3.3 Training and checkpoint selection

Each reference shower determines the visibility, total deposit, layer activity, budgets, channel counts, support, and two flow targets. During training, each component receives the reference values of its upstream quantities. During generation, it receives their sampled values, allowing errors to propagate through the cascade. This distinction between teacher forcing and free-running generation motivates the evaluation of complete showers in Sec. 5. The joint training objective combines nine weighted losses:

$$\begin{aligned} \mathcal{L}_{\text{joint}} = & w_V \mathcal{L}_{\text{BCE}}(V) + w_T \mathcal{L}_{\text{NLL}}(T) + w_F \mathcal{L}_{\text{CE}}(F) + w_A \mathcal{L}_{\text{BCE}}(\mathbf{A}) + w_B \mathcal{L}_{\text{FM}}^B \\ & + w_K \mathcal{L}_{\text{CE}}(\mathbf{K}) + w_S \mathcal{L}_{\text{BCE}}(\mathbf{S}) + w_R \mathcal{L}_{\text{rank}}(\mathbf{S}) + w_Y \mathcal{L}_{\text{FM}}^Y. \end{aligned} \quad (13)$$

Binary cross entropy (BCE) trains visibility, later-layer activity, and channel-support decisions; categorical cross entropy (CE) trains first-layer and count distributions. The total-deposit negative log likelihood (NLL) uses the density in transformed q_T . Expressing that density in deposited-energy units introduces the Jacobian factor $1/(T + 10 \text{ GeV})$. The ranking term averages $\log[1 + \exp(-(s_i - s_j))]$ over sampled occupied–unoccupied channel pairs. Superscripts B and Y denote the masked layer-budget and channel-share flow losses in Eq. (6); the count loss downweights zeros in inactive layers.

The retained calibration proposal uses training-data gradient norms to set w_j . In that proposal, response, layer-profile, and count weights hit the lower clipping bound; the visibility weight hit the upper bound. The bounds apply before normalization of the weights to mean one. Appendix A records the calibration values and their provenance limits. A weight-sensitivity study would test whether these bounds affect shower structure.

Optimization proceeded through response, profile, count, support, share, and joint stages. The condition encoder was trained with the response head, frozen for the intermediate component stages, and unfrozen for joint training. Each stage inherited the preceding checkpoint; freezing the encoder kept the inputs to previously trained heads fixed during intermediate stages.

Training used AdamW with a peak learning rate of 3×10^{-4} , batches of six accumulated over four steps (effective batch size 24), gradient clipping at one, and seed 20260723. Epoch 90 minimizes the teacher-forced validation objective across 104 retained rows spanning epochs 11–114 and defines the checkpoint analyzed below. Selection used this objective; the free-running support diagnostics were evaluated subsequently. The checkpoint need not minimize those free-running discrepancies.

4 Evaluation protocol

We examine the generated readout at three scales: total and section deposits, longitudinal layer structure, and channel-level support. Response measurements include empty showers and are summarized both over the full sample and in eight 25 GeV incident-energy bins. Numerical checks test the generated deposits against the model’s sampled budgets, while distributional comparisons assess agreement with Geant4.

Support measurements use a strictly positive energy definition; no detector or analysis threshold is applied. For a nonempty shower, let F and L be the first and last active layers, and A_ℓ the activity indicator. The occupied span is $L - F + 1$ and the number of inactive layers within it is

$$G = L - F + 1 - \sum_{\ell=0}^{64} A_\ell. \quad (14)$$

This counts inactive layers, not contiguous runs. An interior gap occurs when $G > 0$. We also count active channels and weakly connected components $C_1, \dots, C_m$ of their induced subgraph, ignoring edge direction, and report the largest-component fraction $\max_j |C_j|/|S|$ for occupied set S . A component is a group of occupied channels joined by paths using only occupied channels. These support metrics condition on nonempty readout, leaving 9,907 reference and 9,858 generated showers. The two nonempty subsets need not contain the same incident conditions.

The component count depends on longitudinal activity even if within-layer support were unchanged. Graph edges join only the same or adjacent layers. If R is the number of maximal runs of occupied layers, then $m \geq R$: an empty layer between two occupied runs blocks every path between them. Consecutive empty layers constitute one separation, so G alone specifies neither R nor lateral fragmentation. For example, occupied layers 1, 2, 5 have span five, $G = 2$, and $R = 2$, whereas 1, 3, 5 has the same span and G but $R = 3$. Define $Q = m - R$, the component count beyond one per occupied-layer run. Since $1 \leq R \leq G + 1$, separately averaged nonempty samples obey

$$\Delta \bar{m} - \bar{G}_{\text{gen}} \leq \Delta \bar{Q} \leq \Delta \bar{m} + \bar{G}_{\text{ref}}, \quad \Delta = \text{generator} - \text{reference}.$$

This is a deterministic bound on the retained sample means, not a confidence interval or causal decomposition. Components characterize graph-defined support, not detector-reconstruction clusters [19].

For the all-event active-channel counts, including empty showers, we use the one-dimensional Wasserstein distance W_1 between the reference and generated empirical distributions. It equals the area between their cumulative distributions and has units of channels [7]. Uncertainty is estimated with 1,000 bootstrap resamples, stratified by incident-energy bin and preserving each matched condition pair. The available 95% percentile intervals cover W_1 and the difference in zero-deposit fractions. Intervals for the energy-bin differences, mean longitudinal profile, gaps, and graph components are unavailable. All quoted intervals describe resampling variation within the development bank.

A classifier two-sample test (C2ST) measures how well a classifier distinguishes generated from reference showers; an area under the receiver operating characteristic curve (AUROC) of 0.5 is the chance benchmark [7, 20]. We quote the original battery of three classifier seeds, which uses a row-wise random train/test split. Members of a matched condition pair can enter opposite partitions, so this battery serves as a development screening test. Its condition-only AUROC is 0.464, consistent with a possible opposite-label matched-twin bias rather than a successful control. These row-split scores are not calibrated fidelity estimates. A separate monitor keeps matched pairs within the same partition. On its own event bank, it gives condition-only AUROC 0.500 and high-level AUROC 0.893. Its different event sample precludes a direct comparison between the two batteries. Appendix B gives each classifier seed and the recovered evaluator specification.

5 Results

Table 1 compares deposits over all 10,000 conditions and support over separately nonempty samples. Similar mean deposit and activity coexist with larger longitudinal spans, gap counts and graph-component counts.

Table 1: Aggregate comparison on the development bank. Deposit means and zero-deposit counts use all 10,000 matched conditions; support means use nonempty showers (9,907 Geant4 and 9,858 generated). Differences are generator minus reference, computed before rounding; pp denotes percentage points.

Quantity	Geant4 reference	Generator	Gen. – ref.
Zero-deposit events	93 (0.93%)	142 (1.42%)	+49 (+0.49 pp)
Mean total deposit (GeV)	4.324	4.369	+0.045
Mean ECAL deposit (GeV)	2.100	2.275	+0.175
Mean HCAL deposit (GeV)	2.224	2.094	−0.129
Mean first active layer	0.510	0.735	+0.225
Mean last active layer	56.20	60.62	+4.42
Mean active layers	54.58	54.83	+0.25
Mean occupied span	56.69	60.88	+4.20
Mean inactive layers in span	2.10	6.05	+3.95
Events with an interior gap	52.43%	93.53%	+41.10 pp
Mean active channels	1617.6	1588.8	−28.7
Mean weak graph components	23.42	59.39	+35.97
Mean largest-component fraction	94.90%	88.87%	−6.03 pp

5.1 Deposited energy and empty showers

The all-event mean total deposit is 4.369 GeV for generated showers and 4.324 GeV for Geant4, a difference of 0.045 GeV (1.0%). The similar total means conceal opposing section-level shifts: generated ECAL deposit is higher by 0.175 GeV and HCAL deposit lower by 0.129 GeV (Table 1). Figure 3 shows their mean longitudinal allocation. Its layer profiles average over events and therefore cannot reveal an individual shower’s interior gaps.

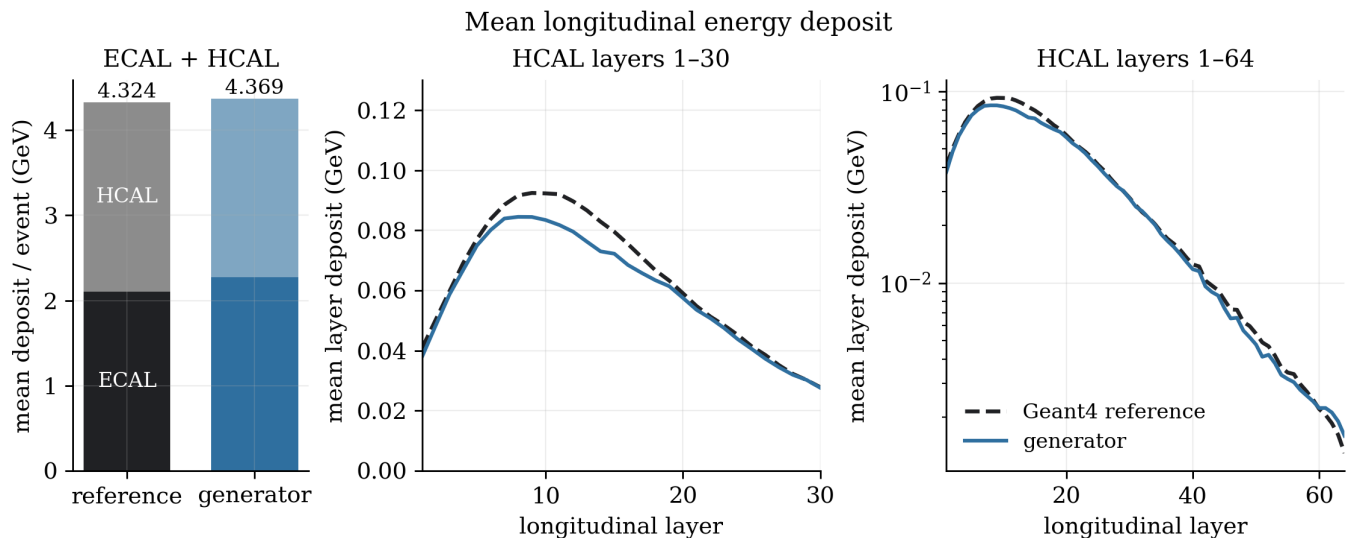


Figure 3: Mean deposited energy over all 10,000 development-bank events. Left: ECAL and HCAL deposits expose section-level compensation. Center: HCAL layers 1–30 (linear scale). Right: HCAL layers 1–64 (log scale). Dashed curves denote Geant4, solid curves the generator; ECAL layer 0 enters only the bars. Event averages include empty showers and do not resolve individual interior gaps.

Across the eight incident-energy bins, generated-minus-reference mean deposits range from -7.7% to $+7.7\%$ of the reference mean; the standard-deviation differences range from -10.9% to $+9.6\%$. These are observed binwise differences; without their sampling intervals, they do not establish energy-dependent bias or distinguish reference fluctuations from model mismatch. Table 3 provides the bin counts, means, and relative differences.

The model produces 142 empty showers versus 93 in Geant4, a difference of 49 events or 0.49 percentage points. The paired-bootstrap 95% percentile interval for this fraction difference is 0.19–0.76 percentage points. The corresponding nonempty samples contain 9,858 generated and 9,907 reference showers.

5.2 Longitudinal activity

The generated showers have nearly the same mean number of active layers as the reference (54.83 versus 54.58), but the occupied layers lie farther downstream. Their mean first active layer shifts by $+0.225$ and last active layer by $+4.42$. The occupied span grows by $+4.20$ layers while the count of active layers grows by only $+0.25$, leaving $+3.95$ inactive layers inside the span. The fraction of nonempty showers with an interior gap rises from 52.43% to 93.53% (Table 1). Figure 4 resolves how much graph fragmentation must remain after occupied-layer runs are accounted for.

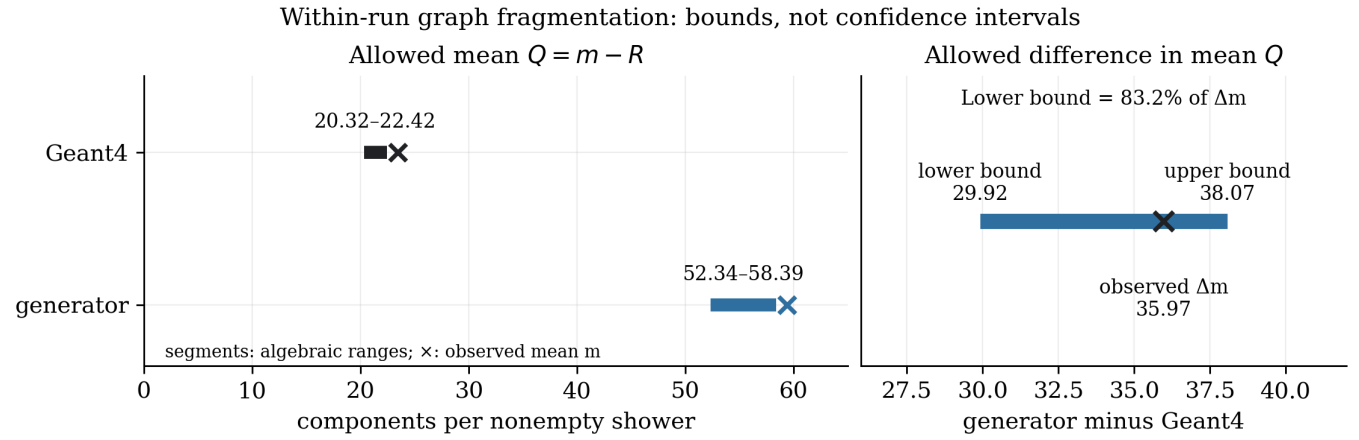


Figure 4: Graph fragmentation conditional on nonempty readout (9,907 Geant4 and 9,858 generated showers). Left: observed mean component counts m (crosses) and allowed ranges of $Q = m - R$ (segments). Right: the allowed difference in mean Q compared with the observed difference in mean m . Ranges follow from $1 \leq R \leq G + 1$; they are algebraic bounds, not uncertainty intervals. All quantities use strictly positive deposits and the model graph.

5.3 Channel support and graph structure

Generated showers have 28.7 fewer active channels on average, but 35.97 more weak components on the model graph. The largest component contains 88.87% of occupied channels on average, versus 94.90% in the reference. The all-event active-channel-count distribution has $W_1 = 58.68$ channels, with a paired-bootstrap 95% interval of 51.47–66.92 channels. This distance includes the zero-count mass, whereas the means above condition on nonempty showers. Two retained reference-half splits give 13.78 and 15.82 channels (5,000 events per half). These are finite-sample comparison scales, not critical values or matched-size significance estimates.

The fraction of directed graph edges whose endpoints are both occupied averages 0.1375 for the generator and 0.1458 for Geant4 over all 10,000 events. This measures local co-occupancy on the same graph, but is not normalized for occupancy and does not isolate a correlation discrepancy.

5.4 Classifier and numerical checks

The original C2ST battery gives mean AUROCs of 0.775 for high-level shower summaries, 0.791 for channel energies and 0.856 for layer profiles across three classifier seeds. Its high-level score exceeds the recorded

screening maximum of 0.65. The failed condition-only control prevents a calibrated fidelity interpretation; a pair-grouped rerun on the same bank is needed. These scores and the failed criterion remain as provenance, not percentages of physically incorrect showers.

The evaluator uses a common event/layer residual tolerance: the larger of 2×10^{-5} GeV and 10^{-5} times the largest sampled event total in each batch. All 1,250 batches of eight passed, with no nonfinite/negative deposit or invalid support/count/mask. Maximum event and layer residuals were 1.14×10^{-5} and 3.05×10^{-5} GeV. The maximum layer residual exceeds the earlier fixed 2×10^{-5} GeV criterion; the reported pass applies specifically to the batch-dependent criterion.

6 Discussion

6.1 Interpretation and diagnostic tests

Graph connectivity distinguishes more than empty-layer separations. The bound in Sec. 4 gives $\overline{Q}_{\text{gen}} \geq 52.34$ and $\overline{Q}_{\text{ref}} \leq 22.42$, hence $\Delta\overline{Q} \geq 29.92$: at least 83.2% of the observed 35.97-component excess remains beyond one component per occupied run. This establishes a within-run graph-fragmentation discrepancy in these samples. It does not identify a purely lateral effect: adjacent occupied layers may also fail to connect through the selected edges. Nor does it assign a causal percentage to a model stage. The separately nonempty populations need not share the same incident conditions.

Equation (4) removes residual activity dependence after conditioning on (h_c, T, F) . As an illustrative example, fix $F = 0$ and let layers 1 and 2 be jointly active with probability p . Independent draws with the same marginals preserve the mean active-layer count but give an interior-gap probability $p(1 - p)$ instead of zero and increase the mean last layer by $p(1 - p)$. This explains a possible signature, not the measured magnitude. Sampled T and F can still induce dependence when their values vary; the first deposited layer is not necessarily the first inelastic-interaction depth.

A reference-derived conditional activity null, controlling incident energy, direction, T and F , would test whether factorization reproduces the observed depth and gap shifts. Conditioning only on reference T and F leaves the factorization intact; an oracle cascade must also replace the full activity mask to test its downstream effect. Layer-conditioned approaches such as L2LFlows and iCaloFlow explicitly model interlayer dependence [21, 22], motivating a matched comparison rather than establishing superiority here.

At channel level, BCE-trained logits are not automatically calibrated inclusion probabilities after fixed-count Gumbel selection. Independent channel perturbations can disrupt connected support even with smooth scores. Occupancy calibration conditional on layer and count, and a matched-marginal placement null, would test that hypothesis. The final channel flow redistributes budgets over already selected channels; it cannot repair support in exact arithmetic.

6.2 Limitations and robustness

Paired sampling intervals for the headline structural means are not retained. Large descriptive differences do not establish robustness across training seeds, thresholds or graphs. The generation hierarchy, small pilot population, teacher forcing and loss calibration remain possible contributors; their effects are not separated by these aggregates.

Strict-positive support is a property of stored simulation deposits, not electronics readout. Soft or late deposits can change gaps and components; their contribution cannot be quantified without channel-energy spectra and time information. Detector-specific threshold and integration-window studies, and an independently defined physical-neighbor graph, are needed before interpreting the graph discrepancy as detector performance. A MIP-equivalent threshold cannot be assigned from the retained metadata alone.

The incident four-vector omits an explicit impact position. Whether direction determines that position depends on the production vertex and recording surface, which need verification. Position-conditioned residuals and the joint empty-shower contingency table would test whether conditioning and geometric acceptance are learned; marginal empty fractions alone cannot answer this.

Training loss in the available implementation is averaged across updating minibatches, whereas validation is measured after the epoch. This can contribute to lower validation loss; the historical runtime match remains unverified. Response-tail studies require counts of generated draws clipped by the cap and reference deposits above it. Numerical sensitivity requires a solver-step study. A locked bank with verified event identities, repeated generation at fixed conditions, and at least three independent final fits are needed for a broader validation claim.

6.3 Computational performance and detector applications

Computational performance is a separate requirement for use as a fast simulator. The available 0.344 s per event is total evaluator wall time, including topology and classifier tests; generation latency was not isolated. A matched end-to-end benchmark must include both flow solvers, decoding, and output handling, with hardware and batch size specified for the model and reference simulation.

Relating this raw-deposit study to detector performance requires production metadata, training-only calibration applied unchanged to both samples, and energy and position reconstruction. The ECAL and HCAL shifts cancel only in the unweighted stored-energy sum; no calibrated-response agreement follows.

7 Conclusion

This pilot generator produces longitudinal and within-run graph-fragmentation discrepancies on a repeatedly inspected validation bank. A difference of 0.25 active layers coexists with a last active layer shifted by 4.42 layers and 3.95 additional inactive layers inside the occupied span. Of the 35.97-component mean excess, at least 29.92 remains after allowing one component per occupied-layer run. The bound rules out empty-layer separations as a sufficient graph description; it does not establish a physical cluster interpretation or a causal model mechanism. Conditional activity and support-sampling tests, together with threshold, timing-window and graph sensitivity, are the next measurements needed to connect this raw-deposit diagnosis to detector performance.

Code and data availability

The [project repository](#) provides aggregate reports, geometry, training history, figure scripts and manuscript source; its README identifies revisions and build audits. The collaboration-owned events and checkpoint are not redistributed. Generation, retraining and event-level studies require those underlying resources. Recorded source hashes establish provenance, not an independently verified match to the historical runtime.

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Author contributions and use of generative AI

Julian Juan: conceptualization, methodology, software, investigation, visualization, and writing.

OpenAI Codex assisted with code and manuscript drafting, reviewing the model specifications, literature searches, and editing. The author supplied the main research ideas, checked the resulting work, and is responsible for the scientific content and final manuscript.

A Recorded settings and reproducibility limits

The analyzed checkpoint is dicos-f-02, epoch 90, from the `calibrated_lr3e4` family, with training seed 20260723. Its SHA-256 is

491284c7423f365230d34b0443f95aa4888ec770bdc673c4c979897bad8acbce

and its recorded frozen-configuration SHA-256 is

116bc8c220b07ce54ae07196bdd6ed8e835775c8c937182a209a799dc94ae9c5

The selected validation objective is 4.4837676194. The recorded lineage includes a declared learning-rate restart between epochs 70 and 71, followed by annealing; it is not one uninterrupted decay. The retained history covers epochs 11–114. The original runtime configuration, complete component-stage durations, optimizer hyperparameters beyond those in Sec. 3, and exact training software environment are not included in the paper package. These details cannot be reconstructed reliably from aggregate losses alone.

Table 2: Recorded gradient-calibration proposal, rounded to six decimals and ordered as in Eq. (13). These values document the retained calibration record; the historical runtime configuration is unavailable for an independent check of the applied weights.

Weight	Component	Value
w_V	Visibility	2.574417
w_T	Total deposit	0.160901
w_F	First active layer	2.159451
w_A	Layer activity	0.536770
w_B	Layer-budget flow	0.160901
w_K	Channel count	0.160901
w_S	Support BCE	1.324108
w_R	Support ranking	1.477591
w_Y	Channel-share flow	0.444960

The retained calibration values are reproduced by taking inverse median gradient norms relative to their geometric mean, clipping to $[0.25, 4]$, and normalizing the resulting nine weights to mean one. This explains why the final entries in Table 2 can fall outside the clipping interval. Gradient norms were measured on training batches with respect to the shared condition encoder. This is a fixed calibration heuristic, not evidence that the weights are optimal.

The diagnostic report records generation seed 20260723 (the same integer as the training seed, used for a separate sampling run), FP32 precision on a CUDA device, batches of eight, and eight updates for each flow. Its validation-manifest identity is

1bc3a6b21f736fc71551580a93207128c043833c6848627fb16f07475931690a

The appendix tables use the same retained aggregate report as the main text; no new shower generation or test evaluation was performed to produce them.

The canonical split is based on event hashes rather than independently identified simulation jobs. The present study uses validation events only. The broader project previously inspected 40,000 nominal-test events in an isolated classifier study and a further draw containing 200 test events, with unresolved overlap. Those historical studies did not feed model decisions, but the nominal test partition cannot be described as wholly untouched; 36,100–36,300 events remain uninspected according to the retained record. A count alone does not identify a locked remainder: exact event identities must be recovered before using it.

B Energy-bin results and classifier details

Table 3 resolves the aggregate response comparison into the eight incident-energy bins. The relative mean difference is $100(\mu_{\text{gen}}/\mu_{\text{ref}} - 1)$ and the width difference is $100(\sigma_{\text{gen}}/\sigma_{\text{ref}} - 1)$, where μ and σ are the mean and standard deviation of total deposited energy. Both statistics include empty showers.

Table 3: Deposited-energy response by incident kinetic energy. n is the number of matched conditions; the same n applies to each sample. Means are in GeV and relative differences are percentages. Bins are lower-inclusive and upper-exclusive, except that the final bin includes 250 GeV (implemented upper edge 250.0001 GeV). Intervals for these differences are unavailable.

K_{inc} (GeV)	n	μ_{ref}	μ_{gen}	Mean diff. (%)	Width diff. (%)
50–75	1,182	2.398	2.216	−7.59	−6.40
75–100	1,270	2.892	2.908	+0.57	+2.80
100–125	1,310	3.527	3.513	−0.42	−0.72
125–150	1,226	3.849	4.146	+7.74	+9.58
150–175	1,288	4.867	4.494	−7.65	−10.89
175–200	1,268	5.030	5.320	+5.76	+8.79
200–225	1,266	5.921	6.026	+1.78	−1.70
225–250	1,190	6.095	6.330	+3.85	+5.24

Table 4: All recorded classifier seeds for the original development screening battery. The three seed columns vary the classifier and its row partition; they are not independent generator-training seeds. All three high-level AUROC’s exceed the recorded screening maximum of 0.65.

Feature family	20260804	20260805	20260806	Mean
Condition only	0.4739	0.4624	0.4545	0.4636
High-level	0.7789	0.7719	0.7735	0.7748
Channel energies	0.7873	0.7943	0.7914	0.7910
Layer profile	0.8490	0.8580	0.8623	0.8564

The archived evaluator implementation uses a histogram gradient-boosting classifier with a maximum of 100 boosting iterations, maximum depth four, and learning rate 0.08. Its class-stratified random split assigns 14,000 of the 20,000 rows to the fitting partition and 6,000 to the scoring partition; the library may further reserve fitting rows for early stopping under its defaults. The same seed controls this split and the classifier. The split does not group matched condition pairs. These are classifier partitions within canonical validation, not the canonical training and test partitions.

The high-level inputs are total deposit, active-channel count, energy-weighted depth and transverse centroids, radial RMS, largest-channel energy fraction, ECAL fraction, and the energy fraction in layers 48–64. The channel-energy family uses all 6,790 deposits, the layer-profile family uses the 65 layer sums, and the condition-only control uses incident kinetic energy. Where a high-level feature divides by total energy, its denominator has a 10^{-9} GeV floor. Other classifier options follow the library defaults; its exact historical version is not established by the retained report.

These settings were recovered from archived implementation source and do not establish an exact source match to the historical evaluation environment. Source hashes and the aggregate report identify the evidence available for this appendix. The 0.65 maximum is recorded in the retained report; its timing relative to analysis is not independently established here. With a 70/30 row split, about $2(0.7)(0.3) = 42\%$ of matched pairs can straddle partitions. Opposite-label twins can bias condition-sensitive scores downward. Table 4 does not replace a pair-grouped evaluation on the same bank.

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